Date : 13/10/2026 **ITA 0329 – Mobile Computing** **C.Eliyazar**

CO1 AT2- Short Answers Test

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1) Spectrum auction = 45MHz
   Company policy sets = 25kHz
   How many full-duplex channels = 45*100/2*25=45000/50=900

2) Given
   Total voice channel =560
   Cell frequency = 7
   Total Channels= 560/7=80

3) Given ,
   Total rolls=100
   Each carrying 8 time slots for voice
   N=Number of carriers*Slots per carrier=100*8=800

4) Given
   Base station power =80
   Hand set ransmit =4
   10 $\log_{10}(80/4)$
   $=10\log_{10}(20)=10(1.3010)=13.01$dB

5) Given
   Test capture signal=-60dBm
   Co-channel interface=-92dBm
   Carrier-to-interface ration(C/I)=-60-(-92)=-60+92=32

6) Given
   Total banddwidth=42MHz
   Channel Bandwidth=300kHz
   Cluster size=7
   N_{total}=42*1000/300=140
   N_{cell}=140/7=20

7) Given
   Initial received power = 64 W
   distance increases by a factor of 2
   path-loss exponent \(n=4\)
   $P2=64(1/2)^4=64/16=4$

9) Given
   Cluster size = 4 cells; each cell has 15 channels.
   N=Cells*Channels per cell=4*15=60

11) Given
   Frame duration = 5 ms; number of slots = 8.
   T=Frame duration/Number of slots=5/8=0.625ms

15) Given
6 cells; each cell handles 250 calls.
$C_{cluster}$ =6*250=1500

16) Given:
Call arrival rate = 4000 calls/second; average call duration = 3 minutes.
N=Call arrival rateAverage duration
Convert minutes into seconds: 3*60=180 seconds
N=4000*180=720000

17) Given
Cahnnels=15
Bandwidth=35
Bandwidth = Number of channels * Bandwidth per channel=150*25=3750kHz

18) Given
No.of Bnads=30
Guard Bandwidth=40
Total guard bandwidth = (Number of bands − 1) * Guard bandwidth
Substitution: (30-1)*40=1160kHz

19) Given
Speed=90
Cell diameter=1.5
Handoffs = Speed ÷ Cell diameter=(90/(2*1.5)=90/3=30

20) Given
Regional Centers=8
Station per center =40
Users per station =25
Capacity = Regional centers * Stations per center *Users per station=8*40*250=80000

21)Given
Total area=400
Cell area=0.4
Number of cells = Total area /Cell area=400/(0.4)^2=400/0.16=2500cells

22)Given
Toatl bandwidth=6000
Cahnnel bandwidth=30
Channels = Total bandwidth / Channel bandwidth=60000/30=200channels

23)Given
Subscribers=1200
Calls perhour =4
Call duration=2.5
Erlangs = Subscribers * Calls per hour * Call duration (minutes) / 60=1200*4*2.5/60=200 Erlangs

24)Given
Total bandwidth=40000
Bandwidth per user =400
Users = Total bandwidth /Bandwidth per user=40000/400=100

25)Given
No.of Stations=1200
User per station =600
Capacity = Number of stations * Users per station=1200*600=720000 users

26) Given
Total cells=200
Cells per round=4
Rounds = Total cells / Cells per round=200/4=50 rounds

28) Given
Total subscribers=600000
Percentage=15
Roaming subscribers = Total subscribers * Percentage ÷/100=600000*15/100=90000

29) Given
Actual load=90000
Maximum capacity=12000
Utilization = Actual load / Maximum capacity * 100=90000/12000*100=75%

30) Given
Total Bandwidth=48000
Channel bandwith=200
Reusefactor=4
Channels per cell = Total bandwidth /Channel bandwidth / Reuse factor=48000/200/4=60

31) Given
Toatl bandwidth=30000
Carrier spacing =200
Channels = Total bandwidth ÷ Carrier spacing=30000/200=150 channels